

DECYRA

AI-Powered Academic & Career Advisor

Requirements Analysis & UX

Problem to Be Solved

Functional and Non-
Functional Requirements

Application Overview

Requirements Analysis



What is DECYRA?

- A digital, personalized mentor for Computer Science students.
- Helps students choose the right Master's program, Erasmus opportunities, and career path.
- Analyzes the user's academic profile and understands their personality to provide tailored guidance.
- Provides recommendations based on a structured database rather than internet trends.
- Enables interactive user engagement through a conversational interface.



Problem DECYRA Solves

Many students make critical academic and career decisions without having a complete picture or access to personalized guidance.

Lack of Clarity

The information available online is often too general and not tailored to individual needs.

Bias

Well-known options receive greater visibility, even though they are not necessarily the best fit for every student.

Lack of Personalized Guidance

There is no unified platform that combines a student's academic profile with their personal interests and preferences to provide customized recommendations.

Critical Decision-Making

During the 3rd and 4th years of study, students must decide whether to pursue a Master's degree, participate in an Erasmus program, or enter the job market—a decision that can shape their entire future career.

Functional Requirements

- User registration and login
- Creation of a personal profile
- Selection of a guidance category
- Interactive conversation with an AI mentor
- Access to a community forum for posting and commenting on experiences
- Chat and video call functionality between users

Non-Functional Requirements

Performance: Fast chatbot response times.

Security: Protection of users' personal data through secure authentication (e.g., Face ID), authentication mechanisms, and a secure database.

Objectivity: Equal consideration and unbiased recommendations across all academic programs.

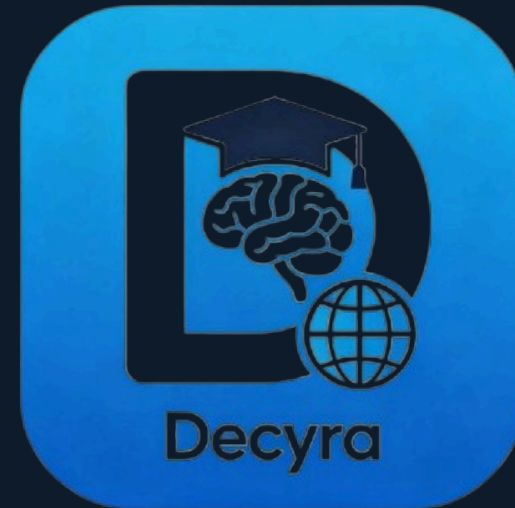
Availability: The system is available 24/7, provided there is an internet connection.

Technologies: Cloud Databases, Protocol usage (etc RTC, HTTP, MCP...), kotlin framework, ML models, AR

Guidance Categories

ERASMUS

- Assistance in selecting the most suitable country, university, and exchange program.
- Comparison of available options based on field of study, language, and budget.



Επαγγελματική Πορεία

- Guidance for entering the IT job market.
- Compare different career options

Master Programs

- Explore Master's degree opportunities in Greece and abroad.
- Filter programs by country, specialization, tuition fees, and duration.

User Experience (UX) Goals

- Fast and effortless completion of the onboarding questionnaire.
- Clear, easy-to-understand recommendations with transparent explanations.
- Users understand why each recommendation is suggested.
- A conversational experience that feels like interacting with a personal mentor rather than a search engine.
- Integration of insights and experiences shared by the community through the forum.
- Interactive communication and exchange of opinions between users.
- A strong sense of privacy and security regarding personal data.

User Profiles & Prioritization

Level 1 — Decision Makers

Third- and fourth-year Computer Science students deciding between a Master's degree, an Erasmus exchange, or entering the job market.

Level 2 — Contributors

Students with experience in Master's programs, Erasmus exchanges, and the DECYRA platform who share their insights and experiences through the DECYRA Forum.

Level 3 — Explorers

Second-year students who are beginning to plan their academic and professional future.



Context Analysis & Data Collection

- **Context:** Students often lack a clear understanding of their available academic and career options and need personalized guidance to make informed decisions.
- **Evaluation Questionnaire:** Real-world data collected from students who represent the target audience and are likely to use the application.
- **Discussions with Classmates and Friends:** Gathered insights into the academic and career decisions students find most challenging.

"It would really help me make such an important decision."

- Third-year Computer Science student

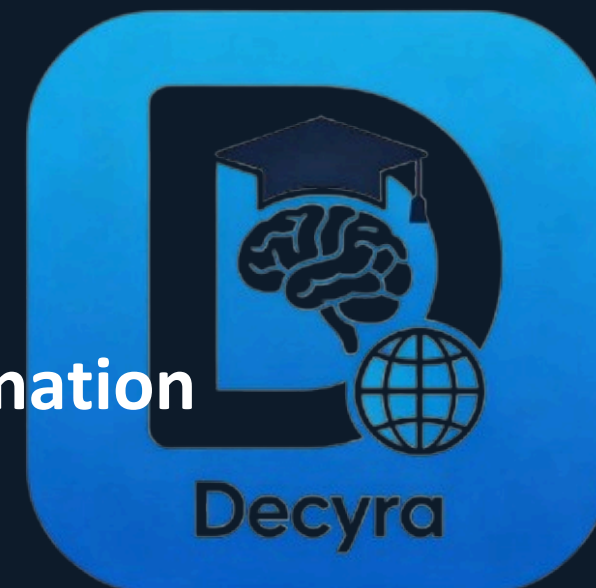
"I wish there were a similar application for Mechanical Engineering—it would have helped me a lot."

- Third-year Mechanical Engineering student, National Technical University of Athens (NTUA)

Research Results

Total Responses: 53

- **The survey was conducted exclusively among 2nd–5th year university students.**
 - **73.6% said they would use an application like DECYRA.**
 - **88.3% reported that finding reliable information about Erasmus opportunities or Master's programs is moderately difficult or difficult.**
 - **92.5% stated that reading other students' experiences would help them make better decisions.**
 - **88.7% said they would write—or would consider writing—their own experiences in the community forum.**
 - **47.2% are not satisfied with the guidance and support services provided by their university.**
 - **37.7% identified cost of living as the most important factor when choosing a destination or program.**



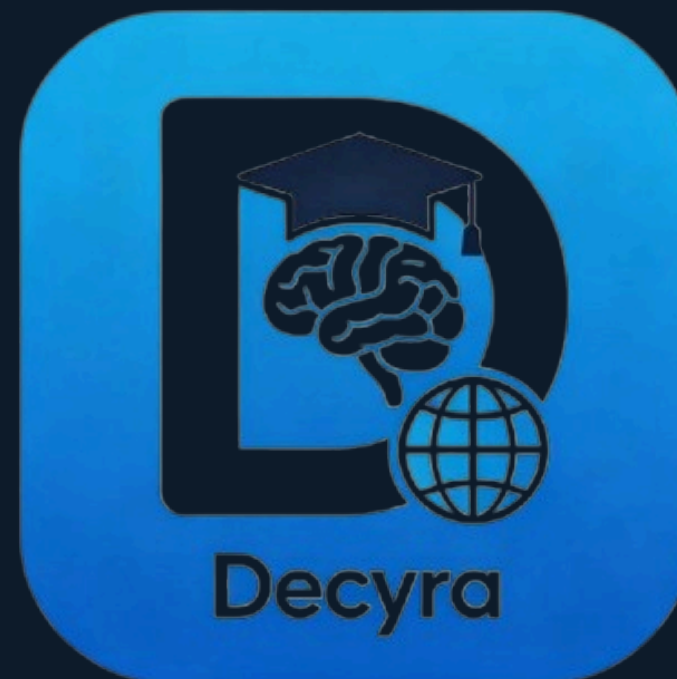
Challenges & Changes from the Initial Idea

Key Challenges

- Ensuring objectivity in results and avoiding influence from popularity or trending choices.
- Proper structuring and design of the database to support reliable recommendations.
- Designing a user-friendly onboarding experience where the mentor gets to know the user through a structured questionnaire.

Changes from the Initial Idea

- Introduction of a Forum for sharing experiences and discussions.
- Addition of Chat and Video Meeting features to enable direct communication between users.



DECYRA as an Essential Tool

- There is currently no structured academic or career guidance system that uses AI to support students in their decision-making process.
- DECYRA combines Artificial Intelligence with direct interaction between users, creating a hybrid model of personalized mentorship and community support.
- The platform ensures a high level of personal data security and privacy protection.
- Completely free of charge, making it accessible to all students.

